Peter Willis — Curriculum Vitae

Senior DevOps Engineer
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Master CV — comprehensive record of all experience, projects, and skills.

Professional Summary

Senior DevOps and Platform Engineer with 20+ years across systems administration, SRE, DevOps, and platform engineering on AWS and Google Cloud. Ships self-service developer platforms, infrastructure-as-code standards, and AI-driven automation that cut lead time, cloud spend, and operational risk for enterprise teams. Career spans hospitality, medical, media, defense-adjacent, and technology industries, split across operations, software engineering, and security analysis for large-scale online businesses. Open-source contributor (Terraform, Packer, GitLab, BusyBox, ProGit); advocate for security, reliability, and Deming-style continuous improvement. Experience leading teams, authoring org-wide RFCs, onboarding acquisitions, and mentoring engineers.

Core Competencies

- **Cloud Platforms:** AWS, Google Cloud (GCP)
- **Infrastructure as Code:** Terraform, Terragrunt, Packer, Ansible, Puppet, CfEngine
- **CI/CD & GitOps:** GitHub Actions, CircleCI, Jenkins, Drone
- **Containers & Orchestration:** Kubernetes, OpenShift, Docker, Helm, ECS, DC/OS, Mesos/Marathon
- **Languages:** Python, Bash, Go, Perl, C, SQL, Java, XML, HTML, CSS
- **Observability & Monitoring:** DataDog, NewRelic, Splunk, Prometheus, Grafana, Ganglia, Cacti
- **Security & Compliance:** AWS SSO, OAuth2/OIDC, Vault, NIST SP 800-171, CMMC v2, DFARS-7020, NMAP, Snort, Wireshark, OpenSSL, SSL/TLS, Symantec DLP, Websense, SAINT Manager
- **Databases & Messaging:** PostgreSQL, MySQL/MariaDB, Redis, MongoDB, DynamoDB, RabbitMQ, Riak, Zookeeper, GraphDB, ElasticSearch
- **Networking & Protocols:** TCP/IP, HTTP, REST, Cisco IOS, all common network protocols
- **Operating Systems & Virtualization:** Linux, Windows, BSD, Solaris, KVM, VMware, Vagrant, OpenStack, Xen
- **Dev Workflow & Collaboration:** Git, JIRA, Confluence, Slack, Teams, Linear, ZenHub, GitHub Flow, Scrum, Kanban, Scrumban, Agile, Lean, Cisco SDLC
- **Leadership:** Team lead (5 engineers), long-term roadmap planning, standups & refinements, RFC authorship, mentorship, cross-team enablement, acquisition onboarding, vendor/account management

Professional Experience

Outside Inc. — Senior DevOps Engineer · Aug 2024 – Present

Maintained the team's long-term roadmap, ran standups and refinements, and partnered with 6 product teams supporting 50 services across 18 AWS accounts.

- Built an automated Terraform repository where deployments re-use DRY modules; users own their own infrastructure through CODEOWNERS grants in a hierarchical tree; GitHub Actions gate merges based on successful plan runs scoped to the actual affected code/modules/deployments to reduce scope of change and number of runs; users apply Terraform by commenting in the PR, and apply only runs if they are a CODEOWNERS maintainer for that specific deployed infra. End result: DRY Terraform, users own their own infra, and can run plan/apply remotely with no local setup and no extra AWS permissions.
- Cut infrastructure lead time from 2 weeks to 2 days by building DRY Terraform modules for developers to roll out new services/infrastructure using only config-file changes (no coding).
- Created deployment pipelines for automated Terraform drift-detection and continuous integration of infrastructure-as-code with open-source tools and GitHub Actions; catches undocumented changes weekly across 18 accounts, 100+ repos, and thousands of resources, saving untold hours of manual redeployment work.
- Built AI-assisted but deterministic solutions for teams to analyze EC2 & RDS usage, identify instances to right-size, evaluate results, approve right-sizing, create tickets for teams, and select Reserved Instances after right-sizing for maximum savings.
- Right-sized RDS and picked standard instance families for best use of reserved-instance credits, driving an estimated 35% AWS cost reduction.
- Triaged and closed several hundred security holes in AWS infrastructure; migrated 100+ human IAM users across 18 AWS accounts (with varying history and permissions/use-cases) to AWS SSO, mitigating decades-old static access keys from being used in attacks, simplifying permissions (from a hodge-podge of random grants to 3 standardized roles), enabling more secure onboarding/offboarding, and letting users switch accounts easily in the console.
- Built a DevOps Slack bot automating common user requests: opening tickets, executing DevOps automated jobs (Terraform, Packer), and initiating whole automated headless AI coding sessions (e.g., to scaffold new Terraform deployments based on standards) and automated AI investigation of incidents using configured skills and tools. Runs in the cloud with Slack modals for ease of use and continuous feedback (checks status and updates the user on progress, even on triggered GitHub Actions).
- Drove 5x more tickets closed per cycle by introducing an "aitask"-labeled Linear workflow: a script detects labeled tickets and fires off headless AI coding sessions (read-only SaaS access, read/write to files in Git) to attempt a pass at the work; the team reviews PRs and applies the result.
- Built automation tools to inventory and update the CMDB, IMDB, and NMDB based on live data from AWS and on-prem resources, helping detect drift, document the system, and save time diagnosing/updating/fixing software and infra while reducing manual errors.
- Developed a plan to rapidly lift-and-shift a vulnerable large-scale business-critical on-prem legacy platform to AWS; executed the migration of ~12 servers and 2 PB of data over ~1 year, reducing risk and improving reliability, security, and performance (including GPU offload for AI/ML workloads).
- Onboarded 5 new acquisitions, migrating/updating legacy infrastructure to modern best practices (~2 dozen VMs, storage, EKS clusters) over ~18 months, including some on-prem moves to AWS.
- Wrote and implemented RFCs for new standard engineering practices in security and CI/CD automation: Git commit signing with 1Password, AWS SSO CLI tools, and developer security analysis tools. All adopted.
- Planned the DevOps team's long-term roadmap, led standups & refinements, and supported developers' needs by customizing Terraform modules and building new infrastructure tailored to their applications.

Tech: AWS · EKS · Terraform · Packer · GitHub Actions · AWS SSO · Python · Slack API · Linear API · Helm · RDS · EC2 · S3

NextEra Mobility — Principal DevOps Engineer · Jun 2022 – Jul 2024

Architected and operated the Google Cloud application platform powering a 30-microservice product.

- Created a standardized workflow for the new Platform group using Jira automation.
- Architected a Google Cloud application platform for improved security of apps and developer access.
- Created IaC in Terraform with CI/CD pipelines to allow devs to change, deploy, and maintain infra easily.
- Created a platform to automate deployment and rollback with CircleCI and custom scripts.
- Cut CI build times 10x, eliminated intermittent failures, and improved security posture by replacing a homegrown Python package repository with Google Cloud Artifacts.
- Improved deploy reliability, minimized build time, and improved caching by replacing a custom VPN with self-hosted CircleCI.
- DRY-ed 80+ repos' CircleCI configs/jobs into one shared library (CircleCI Orb) to improve security and speed up builds.
- Implemented continuous deployment with custom Helm charts on GKE for a 30-microservice product to facilitate easier deploy and rollback; migrated products from old to new infrastructure without downtime using IaC.
- Automated DNS & TLS management with cert-manager and external-dns Kubernetes projects.
- Centralized and automated DNS by migrating from GoDaddy to Google Cloud DNS, managed in Terraform.
- Eliminated shared application credentials, added privilege separation and least privilege; improved security by replacing user service accounts with regular GCP user accounts.
- Triaged and rotated exposed credentials during a major vendor security compromise (CircleCI).
- Automated onboarding to reduce toil and eliminate persistent errors via GitOps (Git/CircleCI) and ChatOps (Opsdroid).
- Implemented cloud-based development in DevSpace/K8s to remove persistent environment/network issues from docker-compose on laptops.
- Researched and implemented a new secure environment for government contracts requiring DFARS-7020, NIST SP 800-171, CMMC v2, etc., to handle CUI; partitioned a copy of the existing product environment into a DOD-secure environment in two months, supporting devs and business users with private-cloud remote desktops.
- Supported product teams with technical issues, helping developers troubleshoot bugs, adapt new technology, and design architecture with cloud best practices.
- Developed custom solutions for SaaS vendors' missing API features in Python and Bash.
- Documented the DevOps practice, employee onboarding, product architecture, vendor/account management, and runbooks for toil; created onboarding and production-support runbooks for developers, business users, and the platform team.

Tech: Google Cloud (GCP) · GKE · Terraform · CircleCI · Helm · DevSpace · external-dns · cert-manager · Opsdroid · Python · Bash · Cloud DNS · Cloud Artifacts

Elsevier, Inc. (Remote) — Senior Systems Engineer · Jun 2019 – Jun 2022

Worked on multiple embedded agile teams, as well as a tiger team across multiple products. Oversaw integration of new acquisition into company engineering standards, and led team of 5 DevOps engineers. Maintained infrastructure-as-code across 5 product families and over 1,000 AWS accounts. Joined via a contract-to-hire engagement through Motion Recruitment, converted to full-time in June 2019.

- Built solutions for multiple teams and products to support Developer and Enterprise needs.
- Provided cloud architecture guidance to developers to meet product goals; built the resulting infrastructure as code using Terraform, Packer, Ansible, and Jenkins.
- Used Terragrunt to manage Terraform across multiple environments; its difficulty keeping configurations DRY and its command workflow not fitting our use cases led me to design terraformsh (see Projects) — a Terraform wrapper implementing a new way of structuring files and running Terraform.
- Led the DevOps Engineer Team in integrating a new acquisition (fastest in business-unit history) into the company S&P; migrated AWS accounts, closed security holes, containerized legacy apps, and adapted the team's legacy SDLC to use Artifactory, SonarQube, Checkmarx, and Contrast.
- Organized cross-team communication channels and documentation hubs for the technology group; wrote user/onboarding guides and runbooks.
- Automated ETL pipelines and a custom SFTP gateway.
- Migrated a live high-traffic production site from a legacy vendor to an entirely custom automated ETL pipeline with a custom-built SFTP gateway in two months, with zero downtime. Managed microservices, MariaDB, SFTP, and web-app infrastructure in AWS protected by CloudFlare.
- Wrote Python and Bash tools to manage multiple environments in IaC monorepos, automating tasks and calling NewRelic APIs to supplant missing Terraform provider support.
- Maintained on-call incident response and wrote runbooks for production infrastructure; debugged production Java and Python services in ECS and OpenShift.
- Designed and implemented new CI and on-call continuous-improvement practices.
- Won 3 company awards for excellence of individual and team engineering achievements.

Tech: AWS · Terraform · Packer · Ansible · Jenkins · ECS · OpenShift · Docker · Artifactory · SonarQube · Checkmarx · Contrast · CloudFlare · Java · Python · Bash · NewRelic

Motion Recruitment Partners, contract to Elsevier, Inc. (Elsevier office, Philadelphia, PA) — DevOps Engineer, contract-to-hire converted to Elsevier · Feb 2019 – Jun 2019

- Implemented disposable AWS ECS clusters using Terraform and Jenkins to build/test/destroy one environment per GitHub Pull Request for faster isolated development/testing. Increased development velocity of the squad by 5x.

Tech: AWS · ECS · Terraform · Jenkins · GitHub

IntePros Consulting (Plymouth Meeting, PA), contract to Comcast Corporation (Philadelphia, PA) — Site Reliability Engineer · Aug 2017 – Aug 2018

- Authored new SRE procedures to unify incident response.
- Added automation that changed deployment from every two weeks to every day.

- Introduced monthly process reviews leading to faster stand-ups, blocker resolution, and timeboxed prioritized work.
- Reduced on-call page-outs up to 6x by converting operational runbooks into autohealing processes orchestrated via Jenkins.
- Responded on-call to incidents and wrote post-mortems for production issues with services used by 50M Comcast customers.
- Shrank tech debt by implementing version-controlled containerized deployments; automated secrets rotation and implemented per-user key management.
- Managed large distributed decentralized Riak clusters on OpenStack and AWS with declarative Ansible playbooks for a large-scale internal service to support all Comcast customers.
- Contributed composable, reusable Python automation tools for autohealing and maintenance on 20% time.

Tech: AWS · OpenStack · Ansible · Docker · Jenkins · Kubernetes · Mesos/Marathon · DC/OS · OpenShift · Riak · Consul · Vault · Confluence · GitHub Enterprise · JIRA · ZenHub · Python

Cisco Systems, Inc. (Columbia, MD) — Automation Engineer III / Software Engineer III · Jan 2013 – Sep 2016

- Built an end-to-end automated test framework for continuous network appliance testing in Perl and C; delivered test features for QA to implement as a matrix of end-to-end feature tests across multiple product lines.
- Delivered test features for QA to implement complex feature-testing solutions across multiple product lines.
- Wrote drivers for the QualiSystems CloudShell automation framework, used by tests to safely share physical test infrastructure across multiple product teams.
- Standardized builds using reproducible infrastructure-as-code with CI/CD and configuration management tools; reimplemented the build process with Ansible, Docker, Jenkins, and Puppet.
- Overhauled the mission-critical performance test suite (Ixia BreakingPoint) for core product lines.
- Collaborated on both an automation development team and a DevOps support team; transitioned from the automation dev team into a DevOps support role.

Tech: Ansible · Docker · Jenkins · Puppet · Cisco Nexus · Ixia BreakingPoint · QualiSystems CloudShell · Matshell · Perl · C · Python

Haymaker Technologies, Inc., contract to University of Maryland Medical System (Columbia, MD) — Information Security Analyst · May 2012 – Nov 2012

- Implemented convergent network security log analysis and vulnerability scanning and reporting.
- Developed tools to scan, identify, and report on site-wide SSL/TLS network service issues.
- Customized search alerts fed by a variety of platform log sources.
- Wrote documentation and reports on a variety of security issues.
- Worked with multiple teams to plan and implement an SSL-inspecting proxy to upgrade an older solution.

Tech: Splunk · SAINT Manager · NMap · OpenSSL · SSL/TLS · Symantec DLP · Websense · Cisco · Juniper · Ansible · Perl

Pencom Systems, Inc., contract to FINRA (Rockville, MD) — Shared Services Engineer · Dec 2010 – Feb 2012

- Supported middleware applications for web application developers.
- Established standards for technology used from development to production.
- Developed tools to gauge realistic system resource estimates on multiple operating systems.
- Planned, documented, and executed multiple upgrades of an IAM solution and dependent components.
- Managed web frameworks and developed tools to assist in administration and migration.
- Assisted in the migration of applications on legacy platforms to new systems in several new datacenters.

Tech: Oracle WebLogic · RSA Access Manager · SpringSource · Perl · SSL/TLS

CBS Interactive (Fort Lauderdale, FL) — Systems Administrator · May 2006 – Sep 2010

- Implemented fail-over functionality for multiple network services.
- Worked with developers to isolate and patch bugs in system libraries and web apps.
- Tracked issues on a web-scale server farm, supporting developers and users of internal services.
- Investigated network and system performance issues on high-traffic production websites.
- Maintained web platforms and DNS, packaged Linux software (RPM), and tuned web servers.
- Developed software for monitoring, configuration, and deployment of infrastructure and network services.
- Assisted a fire-fighting team for peak-traffic days to monitor servers and squash issues before they brought down the site.

Tech: Apache · Ganglia · ISC DHCPD · Perl · Python · RPM · Kickstart · Ant · JWS · Tomcat · CfEngine · Puppet · OpenLDAP

MAKO Surgical Corp. (Fort Lauderdale, FL) — Systems Administrator · Jan 2005 – Jan 2006

- Maintained all network services, storage, and printers.
- Worked with software developers and mechanical engineers to find solutions for technical issues.
- Provided desktop and mobile device support for Windows and Linux users.
- Implemented a custom incremental encrypted off-site backup solution.
- Maintained the corporate website, supported remote VPN, and maintained the T1 connection and PBX.
- Worked with QA to document and verify IT procedures and software used for regulatory certification.

Tech: Active Directory · Apache · Bugzilla · Cisco · CommuniGate Pro · CVS · ISC BIND · MS Office · NetApp

Early Career — Hospitality Industry (2001 – 2005)

Not included in generated resumes unless explicitly requested/opted in.

Windstone Development Intl. LLC — Network Administrator · Jan 2003 – Jan 2005 - General IT problem-solving for a network of clients and servers. - Built a custom app to merge mail labels for large print jobs. - Linux domain controller administration. - Wrote custom web scrapers to collect data for direct marketing.
Comtech Administration Inc. — Perl Developer and Consultant · Jan 2003 – Jan 2005 - Developed OO Perl modules for a site-management CGI application. - Built forensic network security applications.

Windrose Hotel Group LLC — Network Administrator · Jan 2001 – Dec 2002 - Provided networked file access, domain logon, web/mail serving, remote access, disk backup, network monitoring, and mail administration. - Ran a Linux domain controller with Windows clients.

Open Source & Projects

- **Terraformsh** (Feb 2021 – Present) — Wrapper for Terraform providing DRY configurations and workflow automation. Built after Terragrunt proved too difficult to keep DRY and easy to use while at Elsevier — terraformsh implements a different way of structuring Terraform files and running Terraform. github.com/pwillis-els/terraformsh · Design philosophy: github.com/pwillis-els/terraformsh/blob/main/PHILOSOPHY.md (link format pending: consider GitHub short link or GitHub Pages)
- **Network Inventory Lookup Assistant** (Dec 2011 – Present) — Open-source networking utility. github.com/psypete/public-bin
- **EtherDump** (May 2004 – Present) — Open-source networking utility. github.com/psypete/public-bin
- **Open-source contributions** — BusyBox, ProGit, slpkg, GitLab, Packer, Terraform.
- **GitHub** — github.com/peterwwillis (canonical) · github.com/pwillis-els

Certifications & Training

- **CompTIA Network+** — CompTIA
- Training toward **CCNA, CCDA, MCSE**
- Knowledge primarily self-taught, supplemented by on-the-job training and security and programming conferences

Education

- **General Equivalency Diploma (GED)** — Florida, Nov 2001 (received GED to finish school early and immediately begin working in the industry)

Employment Timeline

Period	Role	Employer	Location
Aug 2024 – Present	Senior DevOps Engineer	Outside Inc.	Remote
Jun 2022 – Jul 2024	Principal DevOps Engineer	NextEra Mobility	Remote
Jun 2019 – Jun 2022	Senior Systems Engineer	Elsevier, Inc.	Remote
Feb 2019 – Jun 2019	DevOps Engineer (contract)	Motion Recruitment → Elsevier	Philadelphia, PA (Elsevier office)
Aug 2017 – Aug 2018	Site Reliability Engineer (contract)	IntePros → Comcast	Philadelphia, PA
Oct 2016 – Jul 2017	—	Unemployed	—
Jan 2013 – Sep 2016	Automation Engineer III	Cisco Systems (Sourcefire)	Columbia, MD
May 2012 – Nov 2012	Information Security Analyst (contract)	Haymaker → UMMS	Columbia, MD
Dec 2010 – Feb 2012	Shared Services Engineer	Pencom → FINRA	Rockville, MD
May 2006 – Sep 2010	Systems Administrator	CBS Interactive	Fort Lauderdale, FL
Jan 2005 – Jan 2006	Systems Administrator	MAKO Surgical Corp.	Fort Lauderdale, FL
Jan 2003 – Jan 2005	Network Administrator	Windstone Development Intl. LLC	—
Jan 2003 – Jan 2005	Perl Developer and Consultant	Comtech Administration Inc.	—
Jan 2001 – Dec 2002	Network Administrator	Windrose Hotel Group LLC	—